Concepts of Programming Languages

Memory management

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Outline

- **Why memory management?**
- **Garbage collection**
- **Reference counting**
- **Summary**

Creating Mutable Boxes

- Interpret **`valueExpr`**
  - resulting in a **`value`** and potentially updated **`store`**.
- Get new location in the most recent store.
- Put value into new location and store this location in a box value.
- Return the box value and the updated store.

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  //...
  case NewBox(valueExpr) => {
    val (value, boxStore) = interp(valueExpr, env, store)
    val newLoc = nextLocation
    (Box(newLoc), boxStore + (newLoc -> value))
  }
}
```

We allocate a new memory cell

Adding State Syntactically

- ... will add constructs for defining boxes, for setting/getting the value in a box, for changing the value of a variable, and for sequencing two expressions.

```
<BCFLAE> ::= <num>
           | {+ <SCFLAE> <SCFLAE>}
           | {- <SCFLAE> <SCFLAE>}
           | {with {<id> <SCFLAE>} <SCFLAE>}
           | <id>
           | {fun {<id>} <SCFLAE>}
           | {<SCFLAE> <SCFLAE>}
           | {* <SCFLAE> <SCFLAE>}
           | {if0 <SCFLAE> <SCFL
```

```
           | {newbox <SC
           | {setbox <SCFLAE> <SCFLAE>}
           | {openbox <SCFLAE>}
           | {set <id> <SCFLAE>}
           | {seqn <SCFLAE> <SCFLAE>}
```

No operation for deallocation. What will happen eventually?

```
def whatDoesThisDo(n: Int) : Expr = {
  var v: Expr = Num(17)
  for (i <- 1 to n)
    v = Seqn(NewBox(i), v)
  v
}
```

```
interp(whatDoesThisDo(10))
interp(whatDoesThisDo(100))
interp(whatDoesThisDo(1000))
interp(whatDoesThisDo(10000))
interp(whatDoesThisDo(100000))
interp(whatDoesThisDo(1000000))
```

Quiz: What do these code pieces evaluate to?

Heap overflow: out of memory

Adding State Syntactically

- ... will add constructs for defining boxes, for setting/getting the value in a box, for changing the value of a variable, and for sequencing two expressions.

```
<BCFLAE> ::= <num>
           | {+ <SCFLAE> <SCFLAE>}
           | {- <SCFLAE> <SCFLAE>}
           | {with {<id> <SCFLAE>} <SCFLAE>}
           | <id>
           | {fun {<id>} <SCFLAE>}
           | {<SCFLAE> <SCFLAE>}
           | {* <SCFLAE> <SCFLAE>}
           | {if0 <SCFLAE> <SCF
```

```
           | {newbox <SC
           | {setbox <SC
           | {openbox <SCFLAE>}
           | {set <id> <SCFLAE>}
           | {seqn <SCFLAE> <SCFLAE>}
```

No operation for deallocation.

Quiz: Should we add a deallocation operation?

Manual memory management

- Popular until mid-1990s
- Today: mostly used in C and C++

- Problems:
  - Memory leak: developer forgets to release unused object
  - Heap corruption: developer releases a single object multiple times
  - Dangling pointers: after release, pointers to the released object become dangling
  - Distraction from actual programming task

What happens in a double delete?

15

```
Obj *op = new Obj;
Obj *op2 = op;
delete op;
delete op2; // What happens
```

6

What's the worst that can happen wh
the compiler going to throw an error?

c++ pointers memory-leaks

14

It causes undefined behaviour. Anything can happen. In practice, a runtime crash is probably what I'd expect.

share edit flag

answered Feb 7 '12 at 1:23
Carl Norum
149k • 19 • 272 • 355

6 Random puppy might die as well. – Petr Nov 21 '15 at 9:14

1 There was a known case of a persons monitor lighting on fire due to undefined behavior. – Enn Michael Jul 5 at 10:35

add a comment

17

Undefined behavior. There are no guarantees whatsoever made by the standard. Probably your operating system makes some guarantees, like "you won't corrupt another process", but that doesn't help your program very much.

Your program could crash. Your data could be corrupted. The direct deposit of your next paycheck could instead take 5 million dollars out of your account.

share edit flag

answered Feb 7 '12 at 1:24
Ben Voigt
205k • 21 • 242 • 459

19 Or it could order pizza with your credit card. – Seth Carnegie Feb 7 '12 at 1:29

Automatic memory management

In this lecture, we want to investigate mechanisms for automatic memory management:

- **Reference counting**:
  keeping track of references to an *object*

- **Garbage collection**:
  scan heap for unused objects

Outline

- Why memory management?
- Garbage collection
- Reference counting
- Summary

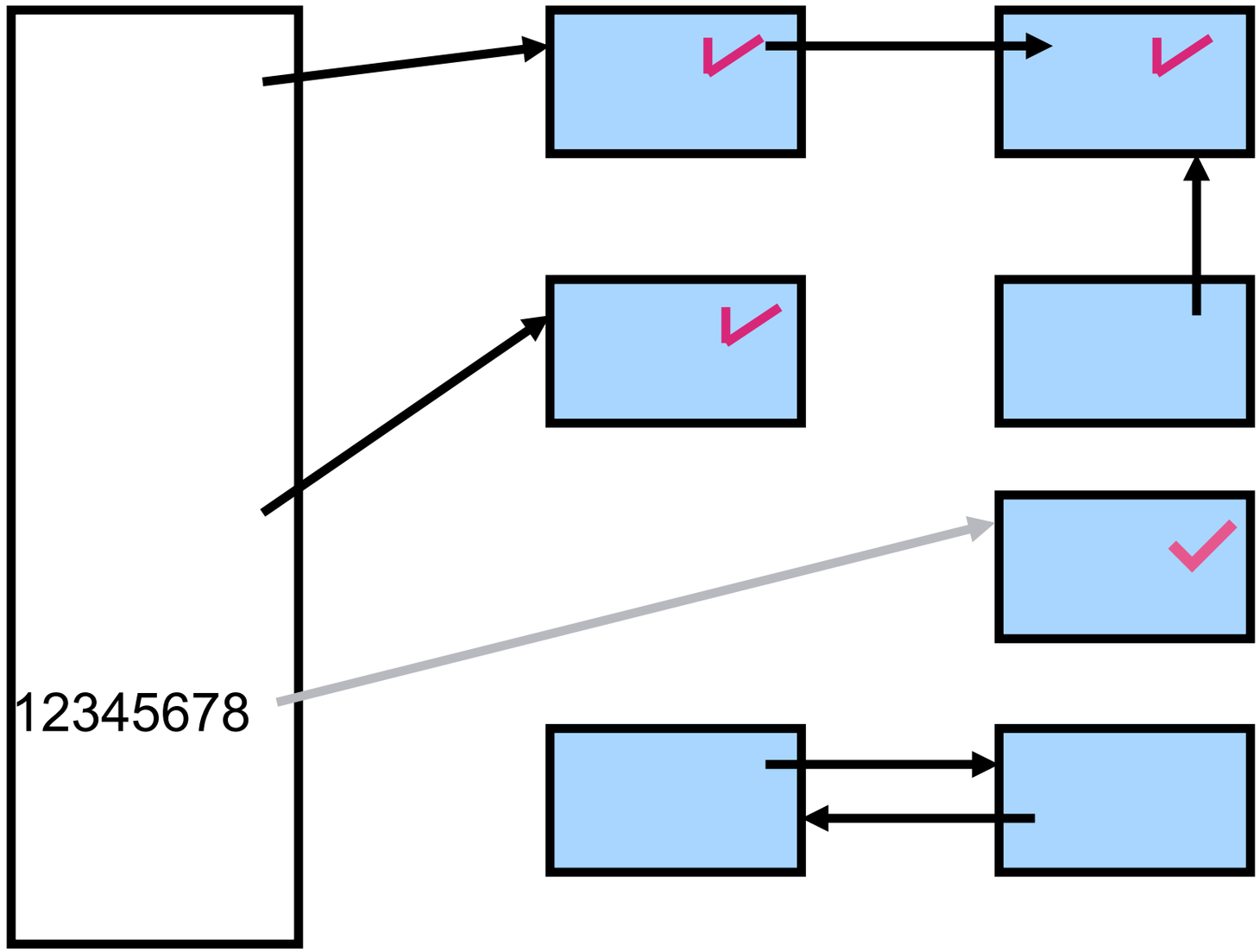
Mark&Sweep garbage collection

Simple idea of Mark&Sweep:

- when memory runs low, scan the heap
- mark objects that are referenced on the stack
  - transitively: mark objects referenced by marked objects
- sweep: release all objects without a mark
- reset the marks

Mark&Sweep garbage collection

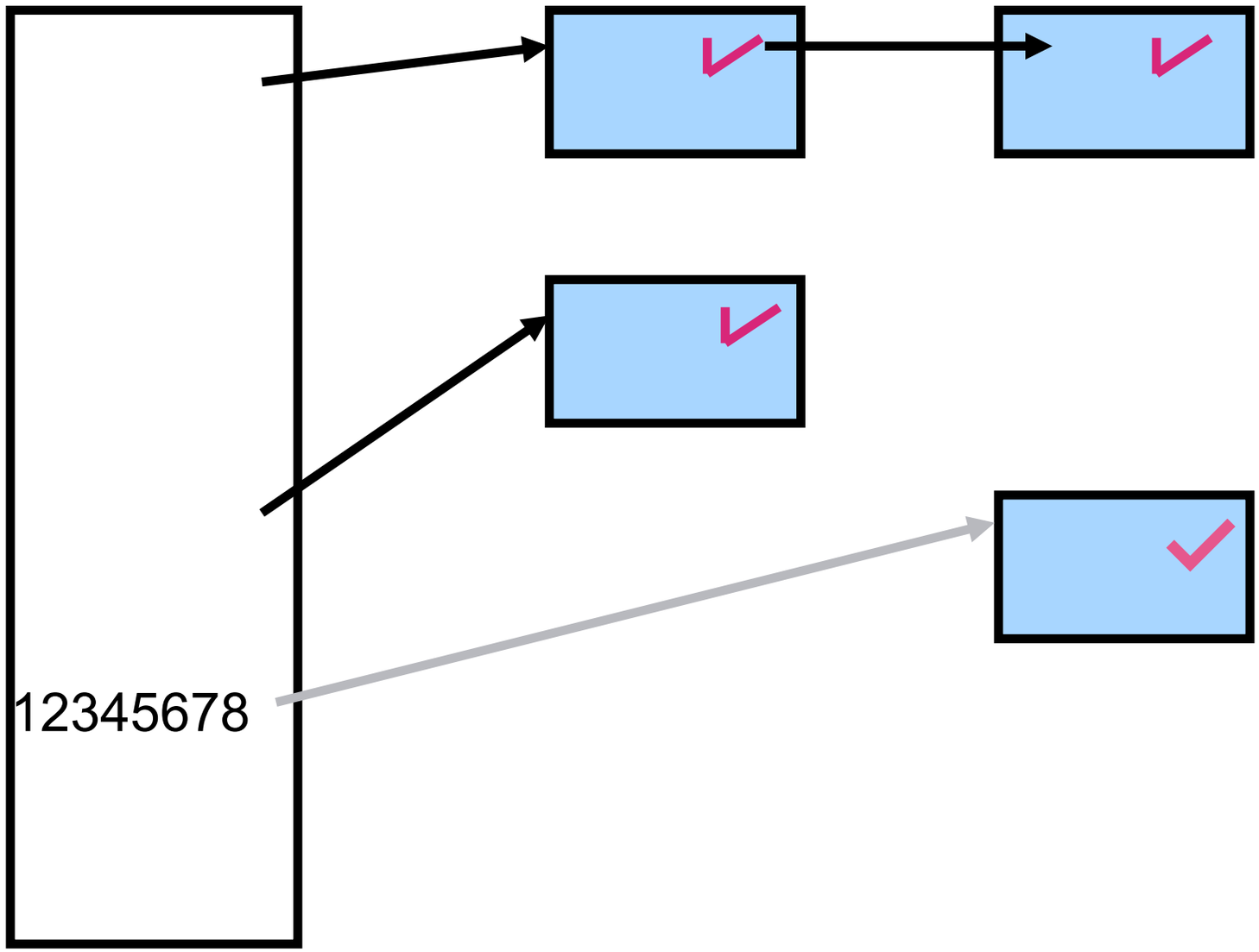
Example:



Stack w/ pointer variables

Mark&Sweep garbage collection

Example:



Stack w/ pointer variables

Mark&Sweep garbage collection

Simple idea of Mark&Sweep:

- when memory runs low, scan the heap
- mark objects that are referenced from the stack
  - transitively: mark objects referenced by marked objects
- sweep: release all objects without a mark
- reset the marks

Let's implement this in our interpreter.

Issues with Mark&Sweep GC

Simple idea of Mark&Sweep:

- when memory runs low, scan the heap
- mark objects that are referenced on the stack
  - transitively: mark objects referenced by marked objects
- sweep: release all objects without a mark
- reset the marks

Quiz: Our GC has some limitations. What are they?

stop-the-world
touches all (virtual) memory pages
heap fragmentation

Other GC techniques

Moving: collect live objects in one part of heap

+ faster allocation, no mark reset for free memory region
- all references must be managed by GC, no pointer arithmetic

Copying: Divide heap into from-space and to-space. When from-space is full, copy only live objects to to-space. Flip roles of spaces.

+ very fast allocation
- can only use half of available memory

Generational: young objects are more likely to die => divide heap into different generations. Run GC on younger generations more frequently.

https://spin.atomicobject.com/visualizing-garbage-collection-algorithms/

Outline

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- Reference counting
- Summary

Reference counting

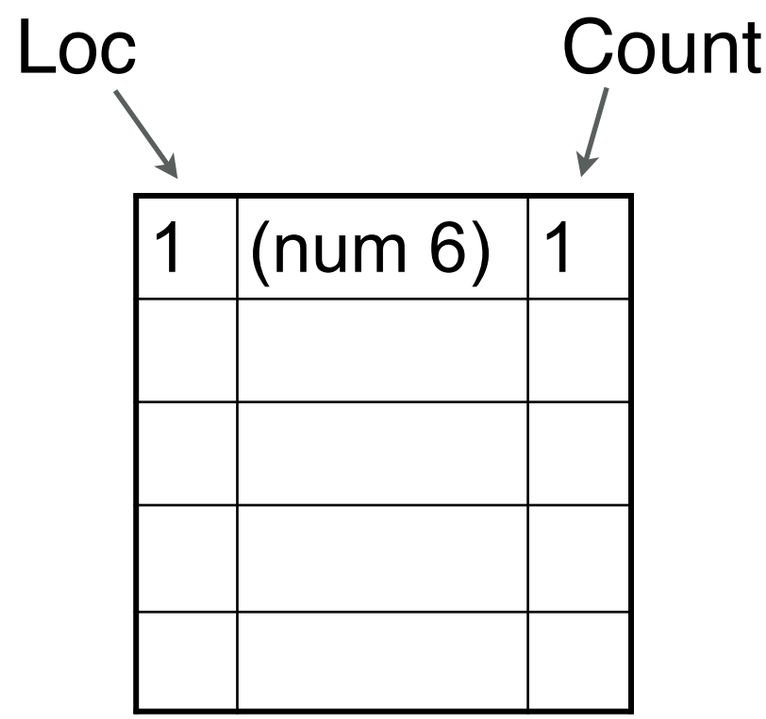
Simple idea:

- Keeping track of number of references for each object
- If the number reaches 0, the object can be released

Reference counting

Example:

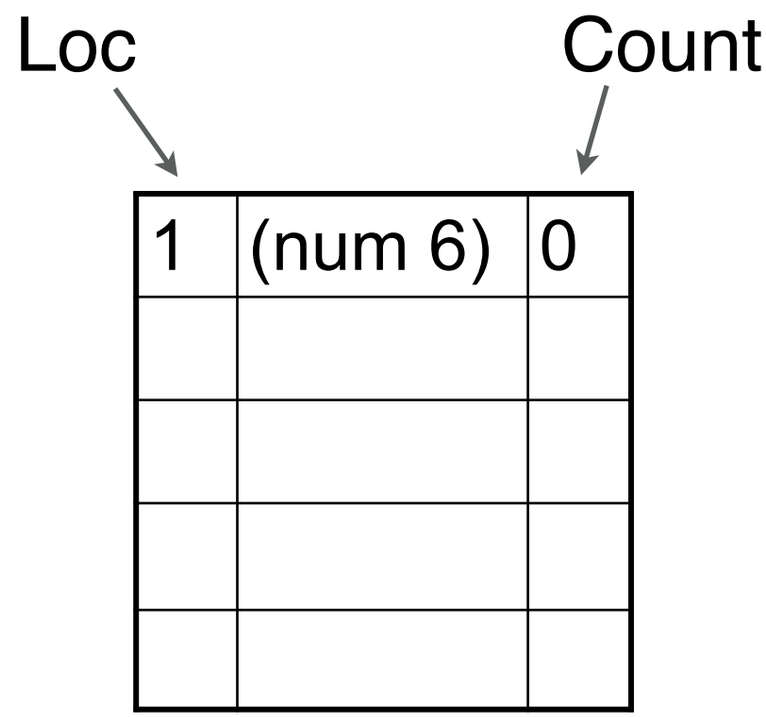
```
NewBox(6)
```



Reference counting

Example:

```
NewBox(6)
```

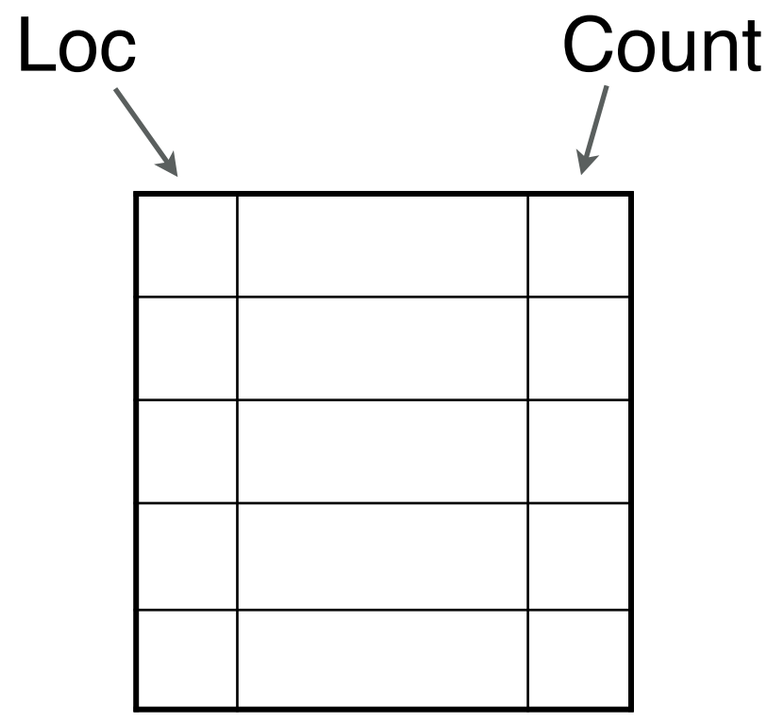


No reference to cell 1 => release

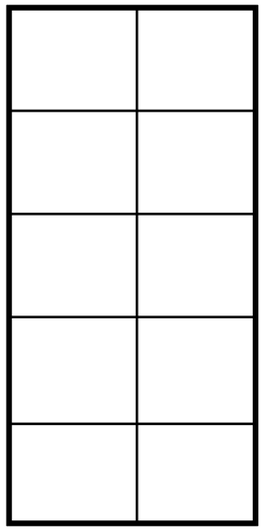
Reference counting

Example:

```
{val x = NewBox(6)
 val y = NewBox(x)}
```



Env



Reference counting

Example:

```
{val x = NewBox(6)
 val y = NewBox(x)}
```

Loc

Count

1	(num 6)	1
2	(box 1)	1

Env

x	2

Reference counting

Example:

```
{val x = NewBox(6)
 val y = NewBox(x)}
```

Loc

Count

1	(num 6)	2
2	(box 1)	1
3	(box 1)	1
4	(box 3)	1

Env

x	2
y	4

Reference counting

Example:

```
{val x = NewBox(6)
 val y = NewBox(x)}
```

Loc

Count

1	(num 6)	2
2	(box 1)	0
3	(box 1)	1
4	(box 3)	0

Env

Reference counting

Example:

```
{val x = NewBox(6)
 val y = NewBox(x)}
```

Loc

Count

1	(num 6)	2
2	(box 1)	0
3	(box 1)	1
4	(box 3)	0

Env

No reference to cell 2 & 4
=> release

Reference counting

Example:

```
{val x = NewBox(6)
 val y = NewBox(x)}
```

Loc

Count

1	(num 6)	1
3	(box 1)	0

Env

No reference to cell 2 & 4

No reference to cell 3

Reference counting

Simple idea:

- keeping track of number of references for each object
- if the number reaches 0, the object can be released

- releasing a single object can free other objects

~~Let's implement this in our interpreter.~~

Issue with reference counting

Simple idea:

- keeping track of number of references for each object
- if the number reaches 0, the object can be released

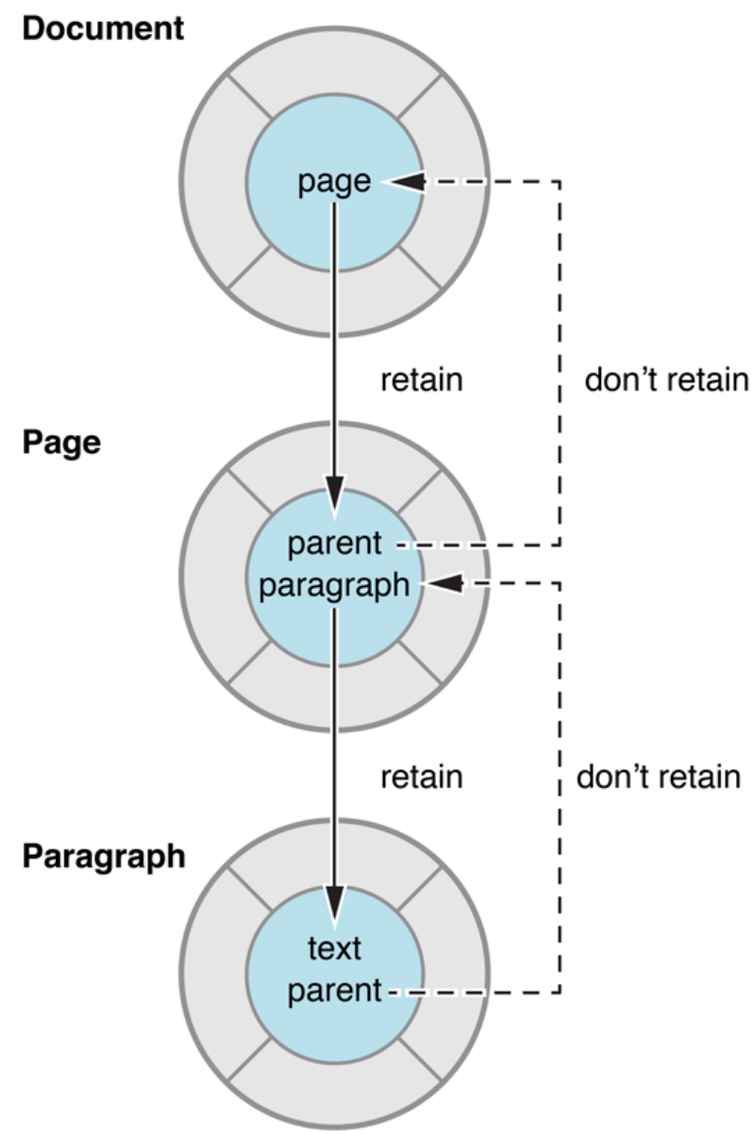
- releasing a single object can free other objects

Quiz: Reference counting has an important limitation. What is it?

- no support for cyclic data structures: reference count never reaches 0

Reference Counting: Handling Cycles

- Design the system to avoid creating them.
  - E.g., forbid reference cycles like in file systems with hard links.



- Weak references
  - "weak" references are non-counted references
  - E.g., in Cocoa recommends using "strong" references for parent-to-child relationships and "weak" references for child-to-parent relationships.

https://developer.apple.com/library/content/documentation/Cocoa/Conceptual/MemoryMgmt/Articles/MemoryMgmt.html#//apple_ref/doc/uid/10000011-SW1

Cocoa (Manual) Reference Counting

```
- (void)takeGroceryList:(GroceryList *)x
{
    // Take ownership
    [x retain];

    // Hold onto a pointer to the object
    myList = x;
}
```

```
- (void)createAndGiveAwayTheGroceryList
{
   // Create a list
   GroceryList *g = [[GroceryList alloc] init];

   // (The retain count of g is 1)

   // Share it with your friend who retains it
   [smartFriend takeGroceryList:g];

   // (The retain count of g is 2)

   // Give up ownership
   [g release];

   // (The retain count of g is 1)
   // But we don't really care here, as this
   // method's responsibility is finished.
}
```

```
- (id)retain
{
    retainCount++;
    return self;
}
- (void)release
{
    retainCount--;
    if (retainCount == 0)
        [self dealloc];
}
```


http://www.informit.com/articles/article.aspx?p=1745735&seqNum=2

Automatic Reference Counting

- Reference counting is accomplished by the programmer.
  - E.g., by sending `retain` and `release` messages to objects
- With **Automatic** Reference Counting these messages are inserted by the compiler as needed.
  - Such feature, based on the Clang compiler, was added in iOS 5and Mac OS X 10.7.

The interpreter uses *automatic* RC.

- Note: retrofitting GC is hard!
  - Mac OS X 10.5 introduced a tracing garbage collector as an alternative to reference counting, but it was deprecated in OS X 10.8.

Outline

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Comparison

Reference counting:

+ incremental: deallocate as soon as object is unreferenced
+ re-use freed cells immediately
- space and time overhead: keeping track of references
- no support for cyclic structures

Mark&sweep GC:

+ handles cycles
+ small space overhead (1 bit per cell)
- requires stopping of regular program execution
- fragmentation: slower allocation, cache misses